LEARNER GUIDE

The alternate segment theorem

Circle Theorems · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **tangent** touches a circle at a point **T**, and a **chord** TB is drawn from that point. The **angle between the tangent and the chord** is equal to the **angle in the alternate segment** — the inscribed angle ∠TPB you see when you look at the same chord TB from a point **P** on the arc on the *other* side of the chord. 'Alternate' just means the segment on the far side of the chord. This is the subtlest theorem in the unit: the two equal angles sit in quite different places. It does not fall straight out of the angle-at-the-centre keystone; instead it is **proved** by drawing the diameter from T and chaining together three facts already known — the tangent meets the radius at 90°, the angle in a semicircle is 90°, and angles in the same segment are equal.

What you are learning to notice

- The **tangent–chord angle equals the angle in the alternate segment** (∠TPB).
- As P slides along its arc, ∠TPB does not change (angles in the same segment).
- The two tangent–chord angles at T (one each side of the chord) add to 180° and match the two different segments.
- The result is built by adding the diameter from T, not by a single rule.

Moving through it, one step at a time

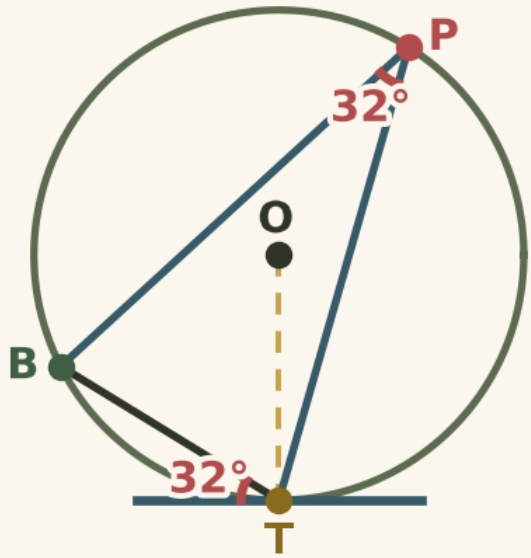
Look for a tangent touching at a point, with a chord leaving that same point — the angle between them is a tangent–chord angle. Its equal partner is the inscribed angle on that same chord, seen from the arc on the other side. Copy the value across and give the reason: 'the tangent–chord angle equals the angle in the alternate segment'. If a diagram gives the far-segment angle, the tangent–chord angle is the same size, and vice versa.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Copy the angle across the chord

A tangent touches a circle at T, and chord TB is drawn. The tangent–chord angle is marked. P is a point on the arc on the other side of TB. State the angle in the alternate segment, ∠TPB, and the reason.

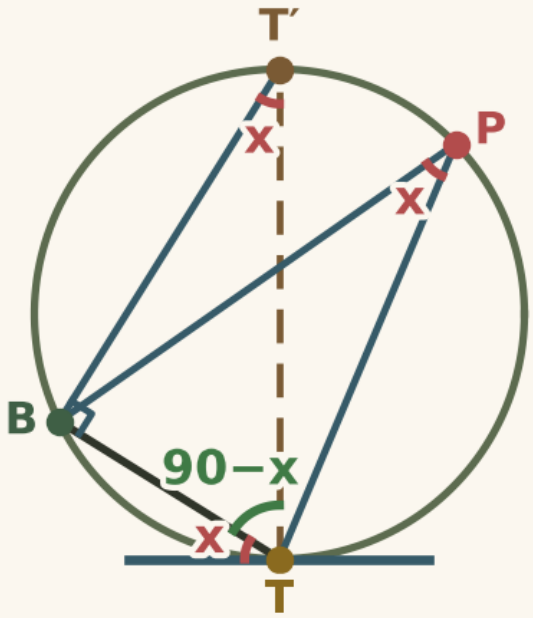


1. Identify the tangent–chord angle at T and the chord TB.	tangent and chord meet at T
2. The angle in the alternate segment stands on the same chord from the far arc.	that is ∠TPB at P
3. The two are equal by the alternate segment theorem.	∠TPB = tangent–chord angle

∠TPB equals the tangent–chord angle — the alternate segment theorem.

Example 2 — Why they are equal (the proof)

Let the tangent–chord angle be x. Prove that the angle in the alternate segment, ∠TPB, is also x.



1. Draw the diameter TT′. The tangent ■ the radius, so the tangent–chord angle is x and ∠(tangent, TT′) = 90°.	tangent ■ radius
2. So the angle between the chord and the diameter is ∠BTT′ = 90 – x.	angles at T
3. TT′ is a diameter, so ∠TBT′ = 90° (angle in a semicircle). In triangle TBT′, ∠TT′B = 180 – 90 – (90 – x) = x.	semicircle right angle
4. ∠TT′B and ∠TPB stand on chord TB from the same segment, so ∠TPB = x.	same segment

∠TPB = x — the angle in the alternate segment equals the tangent–chord angle. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (what a tangent is and that it meets the radius at 90° (Lesson 6), the angle in a semicircle (Lesson 3), and angles in the same segment (Lesson 4)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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